

Building Secure Healthcare AI Assistant Using Groq, Llama3

REVIEW REPORT

Submitted by

21MIS0004 – Ashika.K.R.K

21MIS0017 – Nilani.S.E

for

Natural Language Processing (SWE1017)

PROJECT COMPONENT

Submitted to

Dr. Nancy Victor

**School of Computer Science Engineering and Information
Systems**



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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

ABSTRACT

Chatbots and virtual assistants are set to revolutionize healthcare delivery by transforming traditional patient care into a more personalized, efficient, and proactive process. Chatbots, serving as interactive conversational models, will probably drive this patient-centered transformation in healthcare. In our project, we aim to develop a medical chatbot capable of retrieving patient data and offering basic information after a doctor's consultation. Chatbots, functioning as natural language-based computer programs, form the foundation of this endeavor. Our focus lies in delivering users prompt and accurate data extraction based on their input. Our proposed methodology involves the development of a LLaMA3-based medical chatbot, leveraging cutting-edge deep learning technology to significantly enhance healthcare communication and accessibility. The traditional challenges faced by medical chatbots, such as imprecise understanding of medical conversations, inaccurate responses to jargon, and the inability to offer personalized feedback, are addressed through the utilization of Large Language Model Meta AI (LLaMA).

We are implementing advanced AI assistant designed for private healthcare using the Qdrant Cloud, DSPy, Groq, and Llama3. The Qdrant Cloud ensures scalable and secure data management, while JWT-RBAC enforces robust access control and compliance. DSPy that facilitates sophisticated data analytics and machine learning, enabling personalized patient care. Groq's API acceleration enhances processing speed and efficiency, crucial for real-time healthcare applications. Llama3, a state-of-the-art language model, provides natural language understanding and interaction, supporting complex queries and improving decision-making. This integrated approach creates a secure, efficient, and responsive AI solution tailored to enhance patient management and operational workflows in private healthcare settings.

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1. INTRODUCTION

The emergence of modern technologies like Artificial Intelligence (AI), Internet of Things (IoT) and Natural Language Processing (NLP) has guided into a transformative era in healthcare, offering innovative solutions towards personalized healthcare by enhancing the quality of various medical services. Patient data management in healthcare is the process of collecting, storing, retrieving, and analyzing patient information to deliver high-quality care. It involves gathering data from various sources, securely storing it in electronic health records (EHRs), and quickly retrieving it when needed. Effective patient data management enables accurate diagnosis, personalized care, continuity of care, research, and regulatory compliance.

1.1 OVERVIEW

Patient data requires interoperability, standardization, security, and patient engagement to ensure accurate, complete, and up-to-date data. To provide a robust and secure private healthcare chatbot, revolutionizing technologies like Artificial Intelligence, Machine Learning, Deep Learning and Natural Language Processing are used. In this project using Large Language Model Meta AI (LLaMA) we are creating a chatbot for personalized and enhanced data management. Qdrant Cloud used as the vector database and groq as API provides the needed requirements to develop the Chatbot, while JWT-RBAC provides access control and decides the authentication of users.

1.2 OBJECTIVE

The objective of building a private healthcare AI assistant using Qdrant Hybrid Cloud, DSPy, Groq, and Llama3 is to create a secure, scalable, and efficient AI-powered healthcare platform that:

1. Provides personalized patient care and support
2. Enhances patient engagement and experience
3. Streamlines clinical workflows and decision-making
4. Ensures HIPAA-compliant data storage and processing
5. Supports advanced analytics and research capabilities
6. Improves patient outcomes and population health management

By leveraging Qdrant's hybrid cloud infrastructure, DSPy's distributed computing capabilities, Groq's machine learning acceleration, and Llama3's large language model, this platform aims to revolutionize healthcare delivery, making it more accessible, efficient, and effective.

2. PROJECT RESOURCE REQUIREMENTS

2.1 SOFTWARE REQUIREMENTS

- Qdrant Hybrid Cloud: high-performance vector database designed to efficiently store and search large-scale datasets of numerical vectors
- DSPy: declarative, self-improving framework designed to simplify the development of applications based on Large Language Models (LLMs)
- Groq: software stack for accelerated computing (Groq API)
- Llama3: Pre-trained Llama3 model, environment for deploying language models
- Google Colab: Integrated Development environment that enables python language implementation and cloud integration.

2.2 HARDWARE REQUIREMENTS

- Microprocessor
- CPU
- Cloud infrastructure
- GPU - Graphic processing Unit
- Sufficient RAM
- Workstation with sufficient memory

3. LITERATURE SURVEY TABLE

S. No.	Title	Proposed Methodology	Research Gap
[1]	Network-integrated medical chatbot for enhanced healthcare services. (2024)	the proposed paper uses artificial intelligence and Natural language processing to create a chatbot, an auto-response system. The method involved includes study design, storing data, designing and software. the proposed framework includes three main components, the end-user (manages natural language dialog with the user), Dialog flow (extracts user intents), and fulfillment (supports database).	The common limitation found in these network integrated chatbots are the lack of a mobile application with a simple and modern user interface. And most does not provide reminder for medicine time
[2]	BERT-Based Medical Chatbot: Enhancing Healthcare Communication through Natural Language Understanding (2024)	This paper proposes development of a BERT-based medical chatbot, leveraging cutting-edge deep learning technology. This method includes • Data collection • Text Processing module • BERT model • Context management	Potential biases in training data influence performance and interpretability of complex decision-making processes. Model's effectiveness may vary in handling uncommon medical cases with limited training data

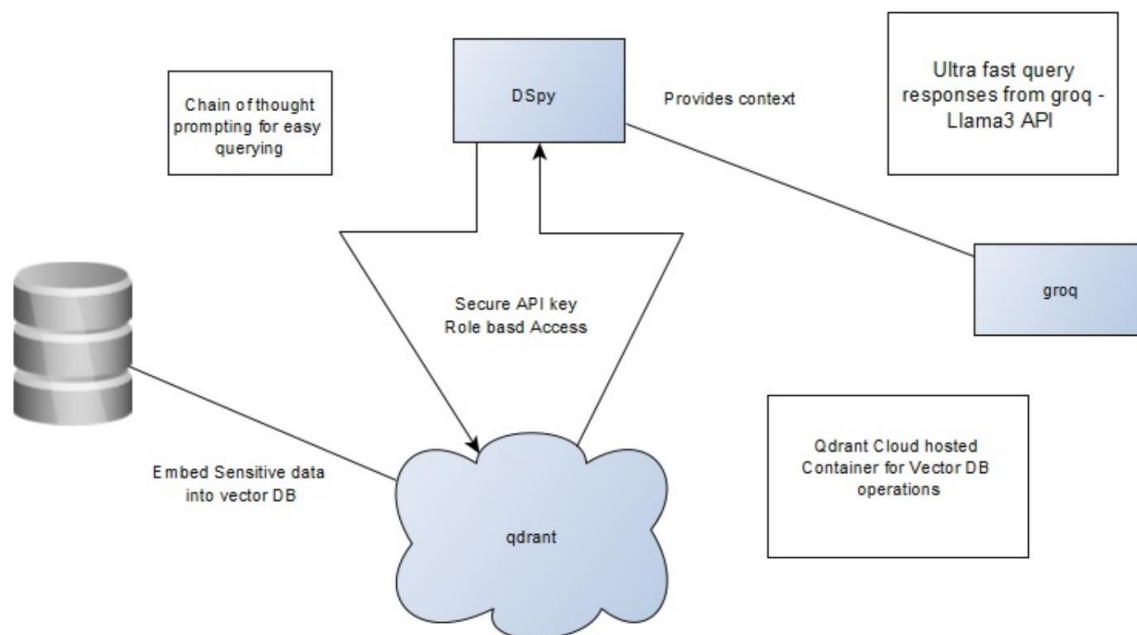
		• Entity and Intent Recognition unit • Dialogue management and response Generation section	
[3]	Foundation metrics for evaluating effectiveness of healthcare conversations powered by generative AI. (2024)	Through the evaluation of Language processing abilities, impact on real-world clinical tasks, and effectiveness in user-interactive conversations, the paper assesses the performance of healthcare chatbots. Also, explores these state-of-the-art LLM based evaluation metrics.	There are several challenges in evaluating these metrics and a few like increase in parameters increase the complexity and affects metrics, subjectivity and the need for a variety of domain expert annotators, modification of parameters can influence chatbot behavior to worst.
[4]	Understanding privacy and security postures of healthcare chatbots. (2022)	Systematic literature review of 1,836 papers and detailed analysis of 40 papers using method including Database Search and Title screening, Abstract screening, Full-text screening and thematic analysis to explore various aspects related to privacy and security concerns in chatbots.	Main limitation found here is the transparency of chatbot with the usage of patient data. Hence, it is difficult to evaluate privacy and security risk as we do not have this information.

[5]	Mental healthcare chatbot based on natural language processing and deep learning approaches: Ted the therapist. (2022)	An AI-based chatbot called Ted that answers questions related to mental health questions with help of natural language processing and deep learning approaches. NLP includes entity recognition program, Intent, Context, Actions and POS. The user message is preprocessed in this step using tokenization, stopword removal, lemmatization, lower-casing and created into Bag of words to be used in deep-learning model. The artificial neural network uses the data prepared to predict the class of user message. Finally, response is chosen and displayed in form of message to user.	The training the health care providers can be extensive. And the empathy and emotion needed to talk with humans might lack in the chatbot
[6]	Multilingual Healthcare Chatbot Using Machine Learning (2021)	This paper proposes a multilingual healthcare chatbot application responding by calculating sentence similarity by using TF-IDF and Cosine Similarity and various NLP techniques.	Lack of data and hence the responses may not be effective. They may struggle with word meaning spelling ambiguities. TF-IDF and cosine

		Random Forest Classifier is used as primary classifier to analyze disease prediction among SVM, KNN, Decision Tree, MNB and Random Forest Classifier.	similarities are limited with dependence on text representations that are language-specific.
[7]	Privacy Concerns in Chatbot Interactions: When to Trust and When to Worry (2021)	This paper proposed a survey on importance of privacy in chatbots and asked user's opinion on • Potential risks that result in concerns in chatbot interaction. • Factors that lead them to trust a chatbot By using proportional-odds logistic regression models to provide an insight on level of trust.	Different age groups may have more significant concerns in their use of chatbots and the survey must enable better disclosures and outcomes for all users The results are not precise and accurate
[8]	A Smart Chatbot Architecture based NLP and Machine Learning for Health Care Assistance (2020)	This paper proposes a survey on existing Chatbots and how they were built using artificial intelligence (AI) technologies, including deep learning, natural language processing, and machine learning (ML) algorithms. This paper also proposed an architecture of a chatbot which includes NLP engine, Intent Classifier,	It is difficult to generate these responses accurately. Large amount of time and data required to handle huge amounts inputs and their responses to be learned.

		Entity Extractor, Feedback Mechanism, Policy Learning	
[9]	Medbot: Conversational artificial intelligence powered chatbot for delivering tele-health after covid-19. (2020)	A conversational Chatbot "Aapka Chikitsak" on Google Cloud Platform for delivering telehealth in India. the main components of proposed work include Dialog flow, Intent and fulfillment. The user Input is designed as query and given to intent that coverts it to code and to the external app, where the query gets solved and appropriate answer is given to intent that is changed into actionable data and shown as output.	The voice assistant is not that well improved to support tests and self-assessments. There is no live connectivity with doctors and booking appointments.
[10]	Chatbot for healthcare system using Artificial Intelligence (2019)	This paper uses the methodology of ranking and sentence similarity calculation are done using n-gram, TF-IDF and cosine similarity	The N-gram and TF – IDF may have problems in understanding medical abbreviations. They measure surface level similarities and hence cannot find similarities for deeper semantic relationships.

4. SOFTWARE ARCHITECTURE



The diagram illustrates the architecture of the chatbot for healthcare system, designed to process and respond to medical queries using natural language. Here's a breakdown of its components:

a. Data Storage (Database):

- This component stores sensitive medical data, such as patient records, medical history, and diagnostic information.
- The data is structured in a way that allows efficient retrieval and analysis.

b. Qdrant (Vector Database):

- Qdrant is a cloud-based vector database, designed to store and manage large amounts of structured and unstructured data.
- In our chatbot, it stores the medical data in a vectorized format, which is suitable for similarity search and AI-powered analysis.
- The data is embedded into the vector database, meaning it is transformed into numerical vectors that represent its semantic meaning.

c. Secure API Key and Role-Based Access:

- This layer ensures that only authorized users can access and interact with the system.
- It likely involves authentication and authorization mechanisms to control who can access the data and perform different actions.

d. Chain of Thought Prompting:

- This refers to a technique where the system breaks down complex queries into a sequence of simpler steps.
- This allows for more accurate and nuanced responses, as the system can reason through the query step-by-step.

e. D-Spy:

- This component likely provides context to the system, helping it understand the medical information and the user's query.
- It uses natural language processing techniques to extract relevant information from the data and the query.

f. Groq:

- Groq is a hardware accelerator designed for AI and machine learning workloads.
- In our chatbot, it provides API for accessing LLaMa.

g. LLaMa 3 API:

- This API provides access to a large language model, potentially trained on medical data.
- The model is used to generate human-like responses to medical queries, drawing on its understanding of medical information.

The system works by receiving a medical query from a user. The query is then processed through the chain of thought prompting and D-Spy to extract relevant information and context. The extracted information is used to search the vector database for relevant medical data. The retrieved data is then processed by the Groq accelerator and the Llama 3 API to generate a comprehensive and informative response to the user's query.

5. MODULE DESCRIPTION

The integration of artificial intelligence (AI) in private healthcare settings offers transformative potential for patient management and operational efficiency. This project accomplishes an AI assistant designed for private healthcare environments, leveraging the Qdrant Hybrid Cloud framework, JWT-RBAC (JSON Web Token-based Role-Based Access Control), DSPy, and Groq with Llama3. Our AI assistant is tailored to enhance patient interaction, streamline data management, and optimize decision-making processes within private healthcare systems.

The Qdrant Hybrid Cloud is a high-performance vector database designed to efficiently store and search large-scale datasets of numerical vectors that provides a scalable and secure platform for managing and processing vast amounts of healthcare data. By combining its capabilities with JWT-RBAC, our system ensures that data access is tightly controlled. JWT-RBAC allows for precise control over user roles and permissions, facilitating secure interactions and protecting sensitive patient information.

DSPy, is a declarative, self-improving framework designed to simplify the development of applications based on Large Language Models (LLMs) which enables sophisticated data analytics and machine learning functionalities within our AI assistant. It supports the development of predictive models and personalized patient care solutions. Groq's hardware acceleration further boosts performance, ensuring rapid processing and real-time responses, which are crucial for healthcare applications.

Llama3, a cutting-edge language model, powers the conversational capabilities of our AI assistant, providing natural and intuitive interactions with both patients and healthcare professionals. This integration allows the assistant to handle complex queries, deliver actionable insights, and support decision-making processes effectively.

Thus, our AI assistant represents a significant advancement in private healthcare technology, offering a comprehensive, secure, and efficient solution for managing patient care and data in a rapidly evolving field.

6. WORK DONE BY EACH MEMBER

TASK	TEAM MEMBER	OUTCOME
Literature Review analysis-5 papers	Nilani	Analyzed the literature to identify methodology and research gap in existing systems
Literature Review analysis-5 papers	Ashika	Analyzed the literature to identify methodology and research gap in existing systems
Team discussion on NLP techniques that are adapted in healthcare industry	Ashika and Nilani	Conducted group discussion that are integrated in the healthcare industry
Identified the problem statement related to the topic	Nilani	Problem Statement: The healthcare AI assistant faces several challenges, including data privacy and security, model accuracy and bias mitigation, real-time response and scalability, and user experience and trust. Thus, we need to, Develop a secure and efficient healthcare AI assistant capable of providing accurate and personalized medical information to patients and healthcare professionals.

Collection of Healthcare Dataset	Ashika	Found out the valid dataset with multiple attributes like patient data, Doctor reports and other health results with more number of rows to train and test the model for accurate question and answering in chatbot
Discussion on model building and development strategies	Ashika and Nilani	Conducted a meeting to discuss the stages of model building and its corresponding techniques
Code Development	Ashika and Nilani	Developed code for a chat bot that works by receiving a medical query from a user.
Requirements installation, Uploading and formatting dataset	Ashika	Installed qdrant-client, groq, sentence-transformers, dspy-ai, fastembed and gradio. Then uploaded Healthcare dataset collected from Kaggle.
Word embeddings using sentence transformer	Nilani	Using sentence transformer generated embeddings in order to store them in a vector DB
Generation of API key from Qdrant cloud	Ashika	Created a cluster collection and uploaded the embedded dataset.
Integration of LLaMa 3 model using Dspy pipeline	Nilani	Groq model is used for acceleration of LLaMa 3 that is used to generate response for given queries.

Development of RAG class	Ashika	Developed Retrieval Augmented Generation class to generate responses using chain of thought prompting and generate answer techniques
Code Implementation	Nilani	Using Gradio UI a chatbot demo is launched.
Calculating Accuracy	Ashika and Nilani	Calculating accuracy score for the responses generated for the respected query using cosine similarity

7. OUTPUT AND SCREENSHOTS

7.1 IMPLEMENTATION SCREENSHOTS

Installed qdrant-client, groq, sentence-transformers, dspai-ai, fastembed and gradio. Then uploaded Healthcare dataset collected from Kaggle. Which are then formatted.

```
!pip install qdrant-client groq sentence-transformers dspai-ai fastembed gradio --upgrade
```

Fig 7.1.1 Requirements Installation

```
[ ] #import files using google colab
from google.colab import files
uploaded = files.upload()

[ ] import pandas as pd
df = pd.read_csv("healthcare_dataset.csv")
```

Choose Files No file chosen Upload widget is only available when the cell has been executed
Saving healthcare_dataset.csv to healthcare_dataset.csv

Fig 7.1.2 Uploading Dataset


```
[ ] # Function to format each row into a single string
def format_row(row):
    return (
        f"Name: {row['Name']}, Age: {row['Age']}, Gender: {row['Gender']}, "
        f"Blood Type: {row['Blood Type']}, Medical Condition: {row['Medical Condition']}, "
        f"Date of Admission: {row['Date of Admission']}, Doctor: {row['Doctor']}, "
        f"Hospital: {row['Hospital']}, Insurance Provider: {row['Insurance Provider']}, "
        f"Billing Amount: {row['Billing Amount']}, Room Number: {row['Room Number']}, "
        f"Admission Type: {row['Admission Type']}, Discharge Date: {row['Discharge Date']}, "
        f"Medication: {row['Medication']}, Test Results: {row['Test Results']}"
        "\n\n".lower()
    )

# Apply the function to each row and create a new column with the formatted text
df['formatted_text'] = df.apply(format_row, axis=1)

# Convert the formatted text into a list (or any other format you need)
text_data = df['formatted_text'].tolist()
```

Fig 7.1.3 Formatting Dataset

Using sentence transformer generated embeddings in order to store them in a vector DB,

```
from sentence_transformers import SentenceTransformer
model = SentenceTransformer("BAAI/bge-large-en-v1.5", device='cuda')
vectors = model.encode(sampled_dataset)
```

Fig 7.1.4 Word Embedding using Sentence Transformer

Created a cluster collection and uploaded the embedded dataset,

```
import os
os.environ['QDRANT__SERVICE__API_KEY'] = 'NHExhUshLcKQIwIWu59rjvNQI90POEbQdDkgc6ffXOF_Q10-yiqE0Q'
os.environ['QDRANT__SERVICE__JWT_RBAC'] = 'true'
```

Fig 7.1.5 Generation of API from Qdrant Cloud

```
[ ] from qdrant_client import QdrantClient
    from qdrant_client.models import Distance, VectorParams

    # Initialize the client

    client = QdrantClient(
        url='https://478dc4ff-e68b-4f91-8bb6-ee83b3d25468.us-east4-0.gcp.cloud.qdrant.io',
        # url='http://localhost:6333',
        api_key=os.environ['QDRANT__SERVICE__API_KEY'],
    )

[ ] client.recreate_collection(
    collection_name="phi_data",
    vectors_config=VectorParams(size=1024, distance=Distance.COSINE),
)

client.upload_collection(
    collection_name="phi_data",
    ids=[i for i in range(len(sampled_dataset))],
    vectors=vectors,
    parallel=4,
    max_retries=3,
)
```

Fig 7.1.6 Creating Cluster and Uploading the Embedded Dataset

Groq model is used for acceleration of LLaMa 3 that is used to generate response for given queries,

```
[ ] import dspy
    llama3 = dspy.GROQ(model='llama3-8b-8192', api_key ="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" )

[ ] dspy.settings.configure(rm=qdrant_retriever_model, lm=llama3)

class GenerateAnswer(dspy.Signature):
    """Answer questions with logical factoid answers."""

    context = dspy.InputField(desc="will contain PHI medical data of patients matched with the query")
    question = dspy.InputField()
    answer = dspy.OutputField(desc="an answer between 10 to 20 words")
```

Fig 7.1.7 Integration of LLaMa 3 model using Dspy

Developed Retrieval Augmented Generation class to generate responses using chain of thought prompting and generate answer techniques,

```
[ ] class RAG(dspy.Module):
    def __init__(self, num_passages=3):
        super().__init__()

        self.retrieve = dspy.Retrieve(k=num_passages)
        self.generate_answer = dspy.ChainOfThought(GenerateAnswer)

    def forward(self, question):
        context = get_context(question)
        prediction = self.generate_answer(context=context, question=question)
        return dspy.Prediction(context=context, answer=prediction.answer)
```

Fig 7.1.8 Development of RAG Class

Using Gradio UI a chatbot demo is launched,

```
[ ] import gradio as gr

with gr.Blocks() as demo:
    chatbot = gr.Chatbot()
    msg = gr.Textbox()
    clear = gr.ClearButton([msg, chatbot])

    def respond(query, chat_history):
        response = uncompiled_rag(query)
        chat_history.append((query, response.answer))
        return "", chat_history

    msg.submit(respond, [msg, chatbot], [msg, chatbot])

/usr/local/lib/python3.10/dist-packages/gradio/components/
warnings.warn(

[ ] demo.launch()
```

Fig 7.1.9 Launching Gradio UI

7.2 OUTPUT SCREENSHOTS

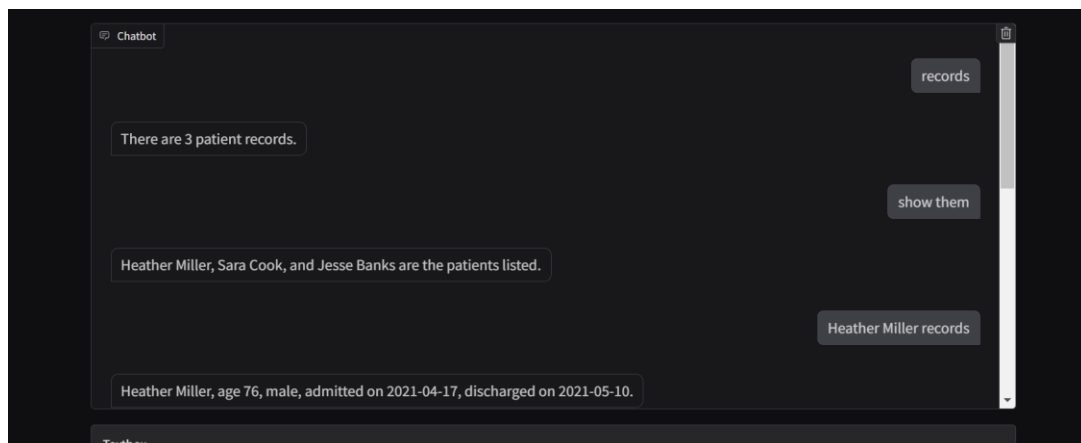


Fig 7.2.1 Output of Chatbot in Gradio UI

The chatbot is providing information about patient records.

Input:

- The user has requested information about patient records.
- The user have previously asked to see specific details about a patient.

Output:

- Number of Patient Records: The chatbot first states that there are 3 patient records.
- Patient Names: It then lists the names of the three patients: Heather Miller, Sara Cook, and Jesse Banks.
- Specific Patient Record: The chatbot displays detailed information about Heather Miller, including their age, gender, admission date, and discharge date.

```
test_queries = generate_queries(df) # Generating test queries
relevance_scores = []
for query in test_queries:
    context = get_context(query) # Retrieve context using your existing function
    relevance = calculate_relevance(query, context) # Calculate automated relevance score
    relevance_scores.append(relevance)

average_relevance = sum(relevance_scores) / len(relevance_scores)
print(f"Average Accuracy Score: {average_relevance}")

Average Accuracy Score: 0.8414303481578826
```

Fig 7.2.2 Finding accuracy using Cosine similarity

8.CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

The integration of artificial intelligence (AI) in private healthcare settings offers transformative potential for patient management and operational efficiency. This project accomplishes an AI assistant designed for private healthcare environments, leveraging the Qdrant Hybrid Cloud framework, DSPy, and Groq with Llama3. Our AI assistant is tailored to enhance patient interaction, streamline data management, and optimize decision-making processes within private healthcare systems. Thus, in conclusion our AI assistant represents a significant advancement in private healthcare technology, offering a comprehensive, secure, and efficient solution for managing patient care and data in a rapidly evolving field.

8.2 FUTURE WORK

In Future, our healthcare chatbot can include integration of Multimodal capabilities, like uploading of images and voice assistant etc., By leveraging behavioral insights, chatbot can offer increasingly personalized and relevant healthcare related advises. Real time data integration ensures up-to-date information on hospitals, patient details and retrieving accurate responses. Improvement on multilingual support and conversational AI capabilities can make this chatbot accessible to global audience. Advanced data privacy and security measures protect patient and hospital information, Ensuring a safe and secure conversation with the chatbot. It can be enhanced through various deep learning techniques to generate precise information on medical records.

9. GOOGLE COLAB LINK

<https://colab.research.google.com/drive/1VsEPrxjx17S1ExNJLT1JE99MYuhLZ5v?usp=sharing>

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